



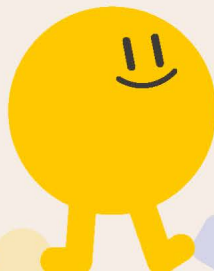
The directive has been issued!

Trolley received!
Instantly running on command!

BLOCK BOT

Author: Yufei Zhang & Nuoqing Shen

Physical programming toy based on color recognition sensors
and remote control cars for children aged 3-6 years.



BACKGROUND

Stages of cognitive development in children and adolescents

-- German psychologist Piaget

0-2 years Sensorimotor stage

Children perceive the world primarily through a variety of bodily processes.

3-6 years Preoperational stage

Children are able to accept simple shapes, but are unable to understand and apply them.

7-11 years Specific operational phases

Children have a rudimentary ability to understand symbols and objects, but not enough understanding of abstract concepts.

12 years+ Operational phase

Children's minds have developed to a level of abstract logical reasoning that allows for hypothetical-deductive reasoning.

Characteristics of children aged 3-6 years



- Children between the ages of 3 and 12 have twice the level of brain activity as adults, and are also at their best for intellectual development.
- Developing children's logical reasoning skills at the age of 3-6 years helps in the subsequent growth of the child.



- Excessive screen exposure (>1h) is not appropriate for children aged 3-6 years old
- Hand muscles are not fully developed in children aged 3-6 years and are not well adapted to fine hand manipulation (use of keyboards, etc.)

There is a need to develop technological toys for children aged 3-6 years old.

RESEARCH

Educational toys for ages 3-6

Features:

Difficult to reason logically, fun and educational
No sharp edges or small parts, safe and non-toxic



Jigsaw Puzzle



Toy Building Blocks

The Need for Programming Education

What society needs:

With the rapid development of China's economy and the increasing popularity of information technology, programming has become an important skill.

Policy support:

The Chinese government has introduced a series of policies in recent years to encourage and support the development of children's programming education. (e.g. the Education Informatization 2.0 Action Plan)



Existing programming platforms for children

Scratch:

A popular graphical programming platform that allows users to create stories, games, and animations by dragging and dropping programming blocks.



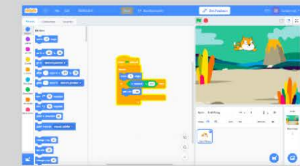
Physical Programming Toys

COMPETITIVE PRODUCT

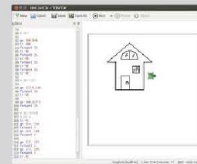
Computer-based and keyboarding operations are not friendly to children aged 3-6 years old

虚拟化

A high degree of freedom and at the same time a high degree of difficulty requires computers, keyboards, microcontrollers, and small parts



Scratch graphical programming language



Logo Programming Language



Arduino programming language

Young age (3+)

Older (12+)



Jigsaw style physical programming toys



Sphero Programming Robot

Small parts, easy for children to accidentally ingest need to be used with a computer

Based on the Scratch programming language to be used with a cell phone

Materialization

Screen-free, free-form block programming toys for children ages 3 to 6 years old.

Note: There are already many products on the market in this category, but they tend to teach programming based on game rules and lack vivid feedback; and those that can do both are often too expensive.

PERSONA



Father

Name: Tom

Age: 34

Occupation: Engineer

Needs:

Hopefully the child will learn some STEM type skills, especially programming

Pain Points:

Programming programs on the market are generally geared toward ages 6 and up, but age-appropriate programming toys often aren't fun enough, or are fun but programming instruction is generally.



Mother

Name: Anna

Age: 33

Occupation: Teacher

Needs:

Would like to purchase some educational toys that are fun and educational.

Pain Points:

Don't want your child to look at a screen for too long and be able to focus on playing for a long time.



Child

Name: Amy

Age: 4

Occupation: Kindergarten

Needs:

Enjoys touching solids with his hands and is interested in cute and brightly colored shapes

Pain Points:

Toys get boring after a while, no high exploration or freedom



Tong Tong
4 years old

I like to play with my tablet, but my mom says I can't look at it for too long, it hurts my eyes.

This is a toy programmed with blocks, you can build blocks to make the car move or complete some tasks, do you think it's fun?

Wow, that sounds so cool! I want the cart to go extra fast and to play with my friends!

PRODUCT

Programming Block

5.4*5.4*4cm



Forward



Back



Turn Left



Turn Right

Parameter Block

*Unit length 6cm



1 Step (6cm forward)



2 Step (12cm forward)



3 Step (18cm forward)

Turn Right



Turn 90 degrees
to the right



Turn 60 degrees
to the right



Turn 45 degrees
to the right

Turn Left



Turn 90 degrees
to the left



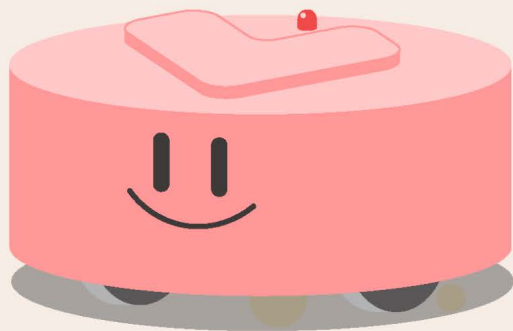
Turn 60 degrees
to the left



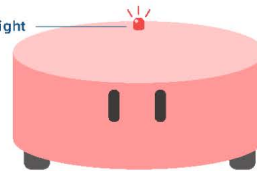
Turn 45 degrees
to the left

Trolley

10*10*6cm



Indicator light



Trolley

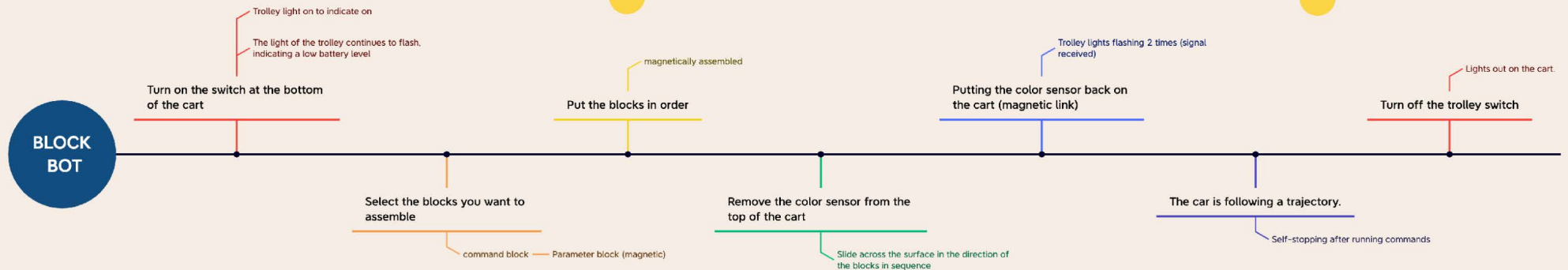
Responsible for receiving
and acting on instructions.



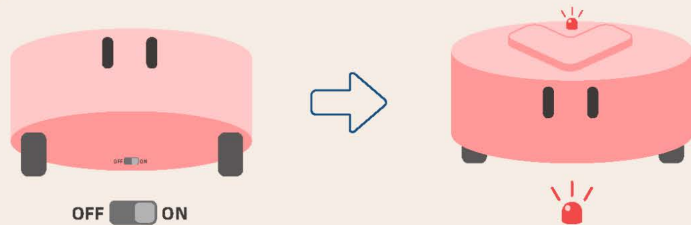
Color Sensor

The color can be read and the instruction
represented by the color can be stored
when sweeping through the programming block.

FLOW CHART



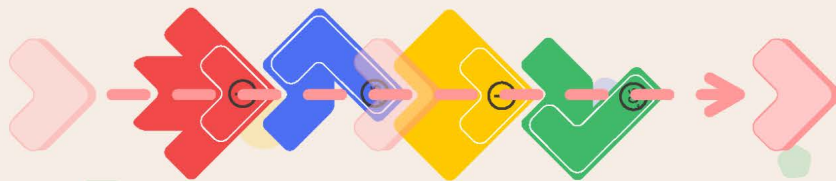
STEP 01 Open the switch



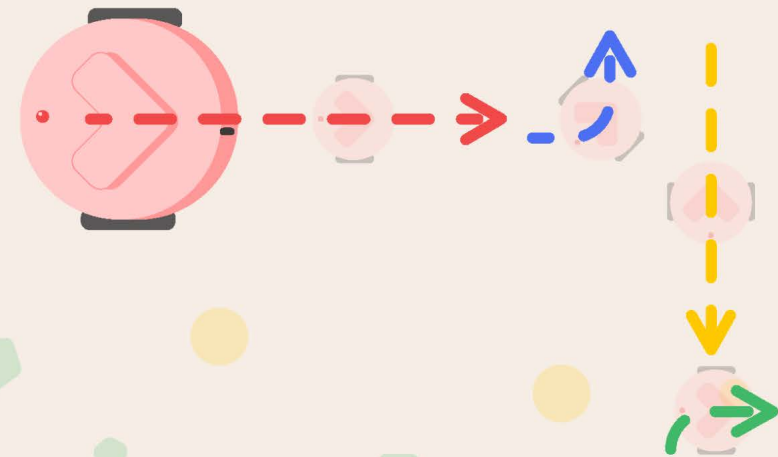
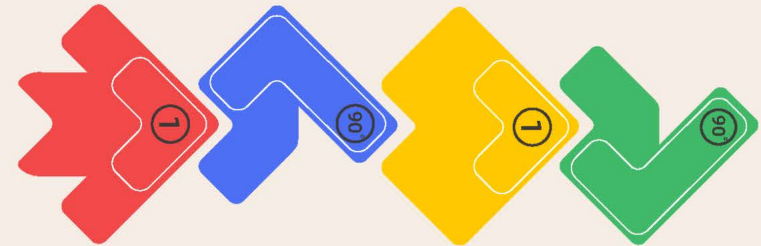
STEP 02 Selection and combination of programming blocks (magnetically connected to each other)



STEP 03 Remove the color sensor and scan



STEP 04 Put the color sensor back into the cart, the cart travels on a trajectory



REALIZATION

* For debugging purposes, the technology is realized with a partial replacement scheme, and the original scheme can be achieved by training models/integration etc. in the future.

Programming Block

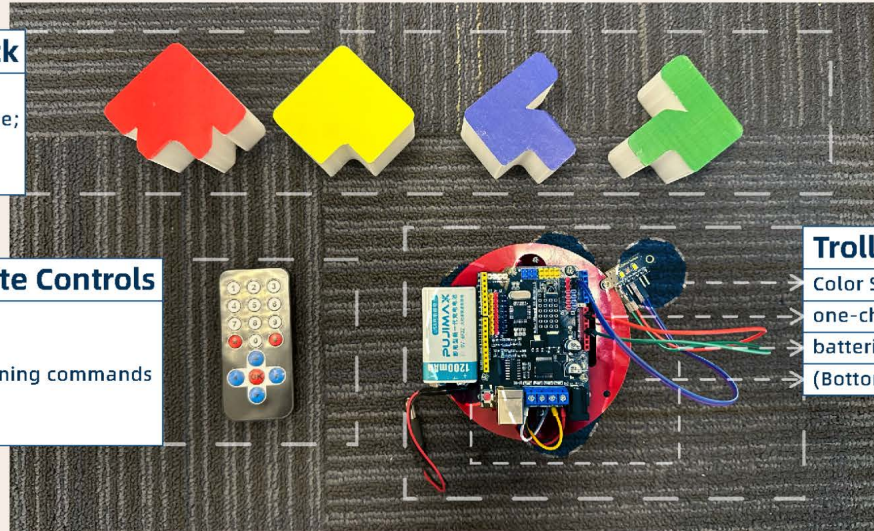
To improve the recognition rate of the color sensor, the model RGB color value is tuned to the maximum value; subsequently, the recognition rate of subtle color differences can be improved by training the model.

Remote Controls

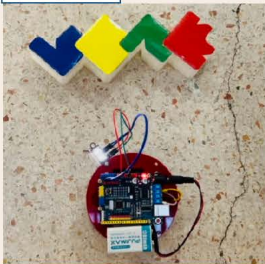
Instruction of the trolley by means of the OK key:
one press: start receiving commands
one more press: end receiving commands and start running commands
(in the original solution this was achieved by removing /installing the magnetic color sensor)

Trolley

Color Sensor
one-chip computer
batteries
(Bottom) Motors, wheels, etc.

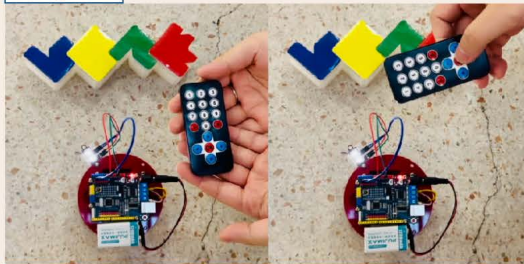


Step 01



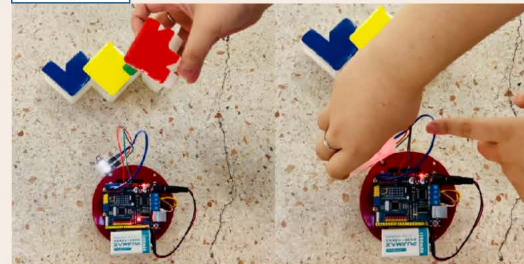
Turn on the switch, the cart light is on to indicate that the power is sufficient

Step 02



Press the OK button, when the small light is on, it means the trolley can start receiving commands.

Step 03



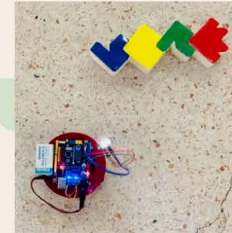
Select the desired programming blocks and put them in order; use the color sensor to scan them in sequence. (For testing purposes, the color sensor here is wired; it can be changed to wireless via Bluetooth technology later on)

Step 04

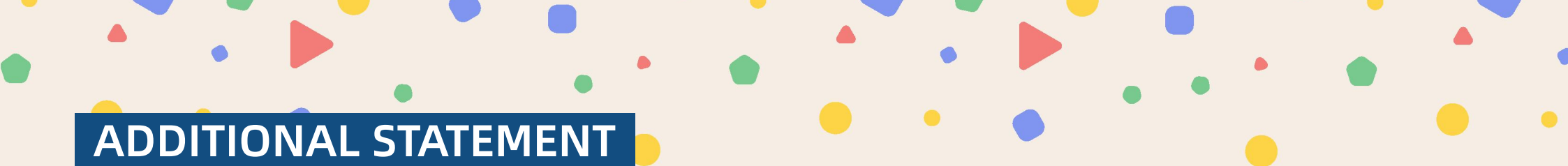


Press the OK button on the remote control and the trolley light blinks once to indicate that a signal has been received.

Step 05



The trolley runs on command and stops by itself after running.



ADDITIONAL STATEMENT

In this project, I was mainly responsible for the implementation of demo. I used KRduino UNO infrared remote control car kit and Tcs3472 color sensor to realize the process of recognizing colors and making the car execute relevant instructions. This is my first time to use arduino programming, and hardware debugging is a very difficult process. Besides, due to the lack of communication technology, I could not retain the infrared remote control mode and color recognition mode at the same time until the end. I hope to strengthen the study of communication knowledge in the future. Demo video and code link: <https://github.com/guaiwu012/BLOCK-BOT>